

Abstract. No. : NIOT-010

Theme : Underwater Communication

TINYML-BASED HYBRID ACOUSTIC AND OPTICAL Li-Fi UNDERWATER COMMUNICATION SYSTEM FOR IoT APPLICATIONS

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Abstract:

This work proposes a hybrid underwater communication framework for IoT applications. The system combines acoustic and optical Li-Fi communication with TinyML-based anomaly detection to improve reliability and energy efficiency.

Introduction:

Reliable underwater communication remains difficult due to energy constraints and unstable long-range transmission. The proposed work integrates hybrid sonar and Li-Fi communication with TinyML-based anomaly detection, to improve transmission speed and energy efficiency. Solar-powered AUVs further enhance sustainable system operation..

Methodology:



Figure 1: Visualization of Hybrid Underwater Communication with TinyML

- Underwater Sensor nodes continuously collect environmental data such as water quality, pollution levels and seismic activity and transmit it to the underwater sink node.
- A TinyML based intelligence model embedded in the sink analyzes the sensor data in real time to detect anomalies and sends alerts to the substation using acoustic communication
- When transferring large size data, high speed Li-Fi communication is used between the sink node and Autonomous Underwater Vehicle(AUV) with equipped with solar panel works as charging the AUVs and also act as receiver in Li-Fi.

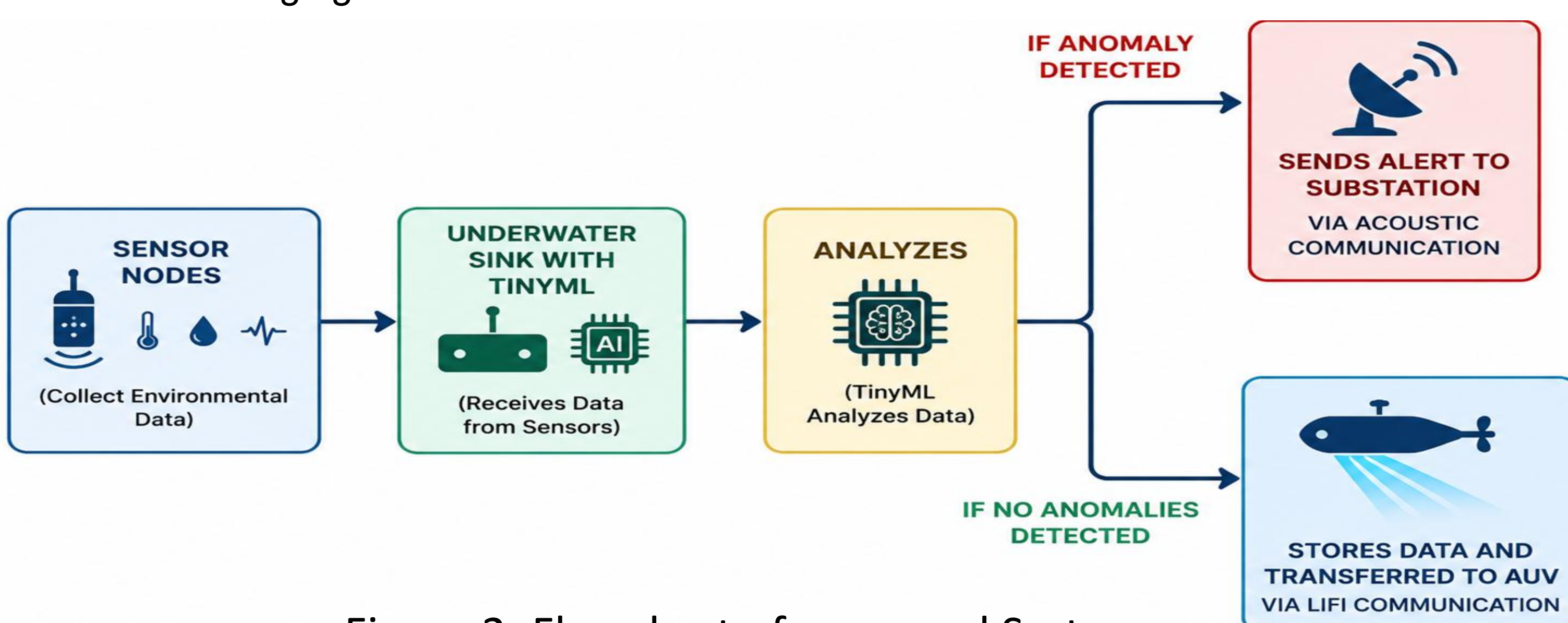


Figure 2: Flowchart of proposed System

Results and Discussion:

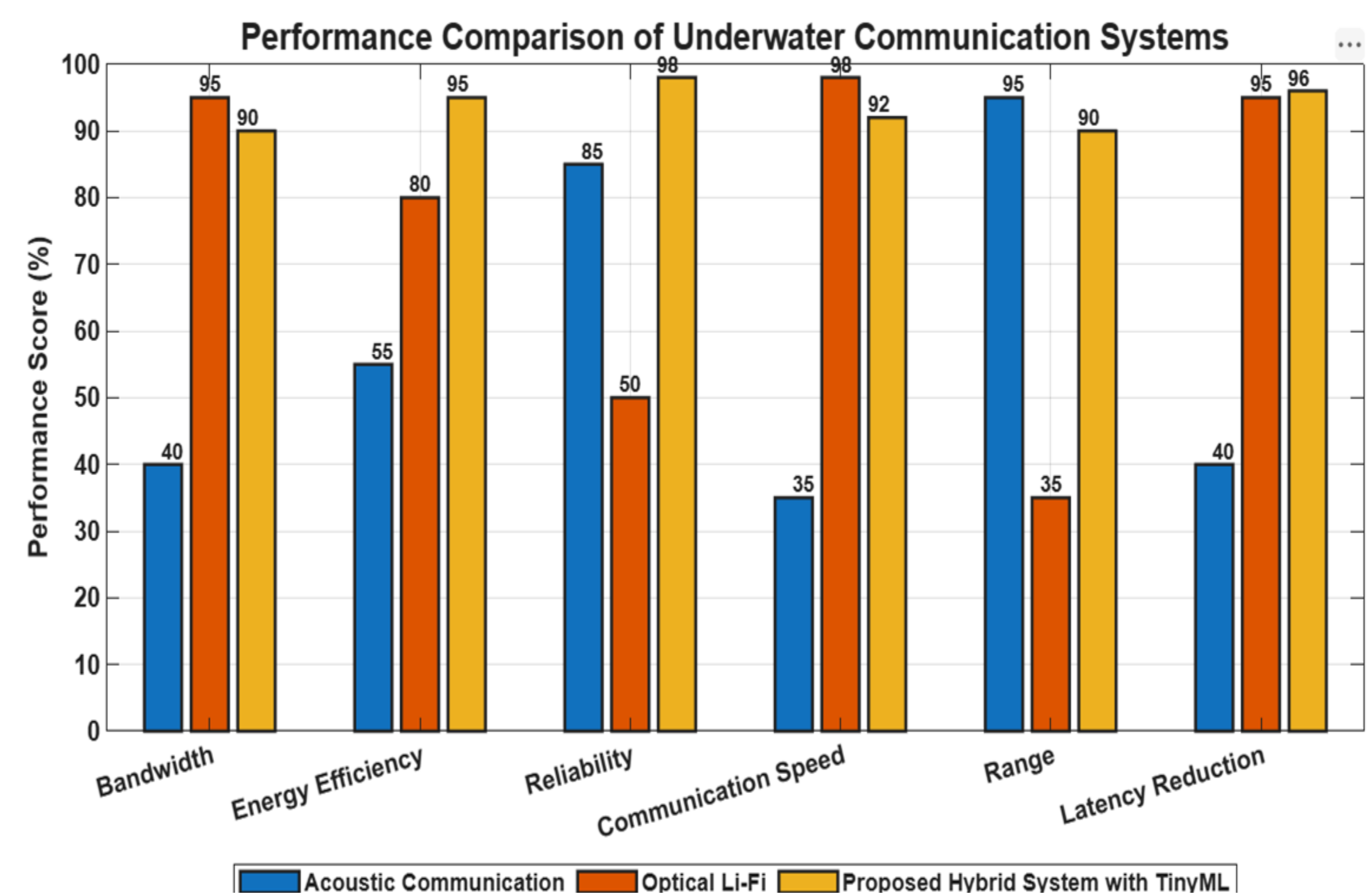


Figure 3: Performance Comparisons of Underwater Communication System

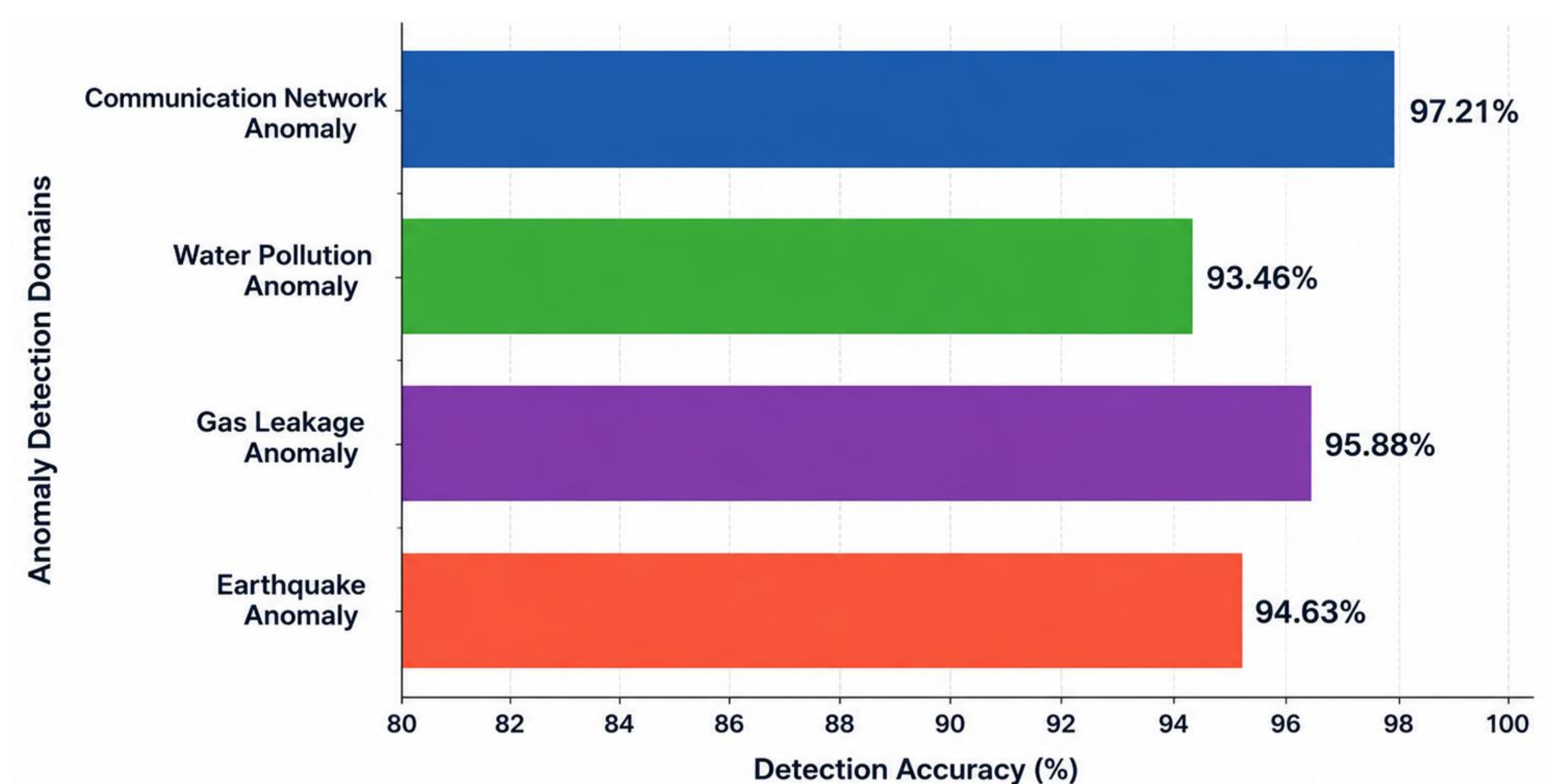


Figure 4: TinyML based anomaly detection accuracy in different domains

- From figure-3 it is clear that the hybrid system provided **high bandwidth (95%)**, **excellent reliability (98%)**, and **reduced latency (96%)**, demonstrating its suitability for real-time underwater IoT applications.
- From figure 4 it is clear that the TinyML anomaly detection showed consistently high accuracy above **93%** in all tested domains, including **communication network anomalies (97.21%)**, **gas leakage detection (95.88%)**, **earthquake anomalies (94.63%)**, and **water pollution anomalies (93.46%)**.
- Experimental results confirm that the proposed framework is effective for **smart underwater surveillance, environmental monitoring, disaster prediction, and secure underwater communication systems**.

Conclusions:

- The proposed **TinyML-based hybrid acoustic and optical Li-Fi communication system** achieved superior overall performance compared to conventional underwater communication methods across all evaluated parameters.
- The integration of TinyML enabled efficient edge-based monitoring and rapid anomaly identification with low computational overhead, improving system responsiveness and reliability in underwater environments.